

Precept 5 - Midterm 1 Review

Key Concepts

- ☐ Definition of a probability measure,
- ☐ Conditional probability,
- ☐ Bayes formula,
- ☐ Definition of the expectation and the variance for discrete random variables,
- ☐ Properties of the expectation (in particular, linearity of expectation without any further assumption, and expectation of a product when variables are independent),
- ☐ Properties of the variance ($\text{Var}(aX + b) = a^2 \text{Var } X$, variance of a sum of random variables)
- ☐ Tower property, Law of total expectation, Law of total probability,
- ☐ Bernoulli Processes and the various distributions covered under this so far (Bernoulli, Binomial, Geometric).

Exercises

Midterm 1 2021 - Q1 (Yogurt)

The dining hall serves (and you eat) one flavor of yogurt each day: blueberry (probability 0.2), banana (probability 0.5), and jalap̃eno (probability 0.3). You have a mild banana allergy: every time you eat banana, you break out in a rash with probability 0.3. The other yogurts do not give you any adverse side effects.

- (a) What is the probability you get a rash on a given day?
- (b) Given that you do not get a rash on a given day, what is the probability the dining hall served banana yogurt that day?
- (c) What is the expected number of times during one week (7 days) that you eat banana yogurt, but do not get a rash?

Midterm 1 2015 - Q3 (Rock, paper, scissors)

This classic two-player game is played as follows. There are three different hand gestures representing **r**ock, **p**aper, or **s**cissors. In every round of the game, each player simultaneously makes one of the gestures (so neither player knows what the other is going to choose). If both players make the same gesture, the round is a tie. Otherwise, one of the players wins according to the following rules: **r** wins against **s**, **p** wins against **r**, and **s** wins against **p**. When the players tie, they keep playing additional rounds until one of them wins, at which point the game ends.

As each player cannot know what the other will play, the best they can do is the following strategy: in each round, each player independently chooses one of the gestures **r**, **p**, **s** with equal probability (that is, each player and each round is independent).

- (a) What is the probability that the first round is a tie?
- (b) What is the expected number of rounds played until someone wins?
- (c) What is the probability that the first player eventually wins the game?
- (d) What is the expected number of times the first player plays **r** throughout the game?

Midterm 1 2019 - Q1 (Questions on the exam) - *time permitting*

The exam has three questions. They are in increasing order of difficulty: you get the first question right with probability $3/4$; the second question right with probability $1/2$; and the third question right with probability $1/4$. Your success in answering each question is independent.

- (a) What is the expected number of questions you get right?
- (b) What is the variance of the number of questions you get right?
- (c) Given that you got two out of three questions right, what is the probability that you answered question number three correctly?